

graphical network monitoring tool written in PHP/MySQL. It makes use of the RRDTool (Round Robin Database Tool) to store data, generate graphics and collect network traffic data using the Net-SNMP protocol. Being a powerful network monitoring tool, Cacti allows systems administrators to collect data from almost any sort of network hardware like routers, switches, firewalls, load balancing equipment as well as servers, and presents the data in properly visualised graphs.

The front-end of Cacti can handle multiple users, each with their respective graph sets, and is mostly used by Web hosting providers to monitor the bandwidth statistics of customers. The back-end of Cacti has two forms: *cmd.php*—a PHP based executable script for smaller installations, or *Spine*—a C-based poller that can scale to thousands of hosts.

The operation of the Cacti Web based monitoring tool is divided into three different tasks, which are described below.

Data retrieval: Cacti makes use of a poller to retrieve data. Its application is executed at regular intervals of time under varied OSs to monitor routers, switches, servers and other network hardware. Cacti makes use of the SNMP protocol for live monitoring of data from various devices.

Data storage: Cacti makes use of the RRDTool to store data either in a SQL or flat database. RRD is a system to store, and shows time series data collected from different SNMP-capable devices.

Data presentation: Cacti has an inbuilt graph presentation based utility to display graphs as per the reports based on the time series data collected from various network devices. Graphs, in turn, provide fast and easy visualisation of data for network administrators to maintain the health of the network 24x7.

The latest version of Cacti is 0.8.8h, and was released in May 2016. Its features are listed below:

- Unlimited graph items, graph data manipulation and graph templates
- Built-in SNMP support, user based management and security
- Data source templates and host templates
- Data gathering on a non-standard time span
- Fully flexible and dynamic data sources

OpenNMS

OpenNMS is regarded as an enterprise grade free and open source network monitoring and management platform for systems and network administrators. It was developed to create a pure, distributed, scalable management application platform for all aspects of network management with special focus on fault and network performance management.

Open NMS provides automated and directed discovery and provisioning, event and notification management, service assurance and performance measurement.

OpenNMS is built using the Java programming language and is available for free under GNU version 3. The OpenNMS package provides us with a complete network management solution which can scale up to thousands of nodes to easily and effectively collect and store network information. OpenNMS enables network

administrators to monitor all sorts of resources, quotas, network usage statistics, etc. Data can be further analysed via graphs, and OpenNMS provides a proper Web user interface for all sorts of data related to network devices. This highly dynamic and flexible tool enables systems administrators to customise dashboards, duty schedules and on-call calendars on a per-user or per-group basis.

The current version of OpenNMS is 18.0, which was released in May 2016.

Its features are:

- **Event management and notification:** OpenNMS is based on the principle of ‘Publish and subscribe’. Processes in the software can publish events and other processes can subscribe to them.
- **Discovery and provisioning:** OpenNMS consists of an advanced provisioning system for adding devices to the management system by submitting the range of IP addresses to the system. It consists of adopters to integrate with other processes within the application as well as external software like a dynamic DNS server and RANCID.
- **Server monitoring:** OpenNMS monitors network based services ranging from very simple ICMP pings to complex protocols like SMTP or page sequence monitoring.
- **Data collection:** OpenNMS collects information of various protocols like SNMP, HTTP, JMX, XMP, XML, NSClient and JDBC.

Icinga

Icinga is a free, open source, scalable and extensible network monitoring application which checks the availability of resources, notifies users of outages and provides extensive business intelligence data. Its new features include a Web 2.0 style user interface, additional database connectors for MySQL, Oracle and PostgreSQL, and a REST API that lets administrators integrate various extensions without modifying the Icinga Core.

The latest release of Icinga is version 2.4.9, which came out in May 2016.

Architecture: Icinga Core is developed in C language and has a modular architecture with a standalone core, user interface and database on which users can install various plugins and add-ons.

The components of the architecture are:

1. **Icinga Core:** This manages all sorts of monitoring tasks and receives various results from plugins. The core communicates the results to IDODB through the IDOMOD interface and the IDO2DB service daemon over SSL encrypted TCP sockets.
2. **Icinga 2:** This manages monitoring tasks, running checks and the sending of all sorts of alert notifications. It can be enabled on-demand, such as the ‘checker’ or ‘notification’ component.
3. **User interfaces:** Icinga has two types of user interfaces.
 - (a) **Icinga Classical UI:** This is based on Nagios CGIs and has new features added to this interface such as pagination, JSON output and CSV export.